Splats on Set: Mapping the Adoption of 3D Gaussian Splatting along the Virtual Production Pipeline

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Abstract

3D Gaussian Splatting (3DGS) can turn normal camera footage into photoreal 3D scenes that run in real time, which makes it interesting for Virtual Production (VP). However, it is unclear where in the VP pipeline it is being used. This paper maps 13 sources (papers, talks and case studies) onto different phases of the VP pipeline and compares them with a small survey of 13 practitioners. Both point in the same direction: Splats are already used and trusted in pre-production, for example for previs and scouting, but hardly as final-pixel content on the LED wall yet. Relighting, moving content and artifacts hold them back. The survey adds one problem the papers barely mention: once captured, splats are hard to edit and art-direct. Final pixel only works for distant, static backgrounds that are easy to relight. Still, all 13 respondents expect 3DGS to become standard in VP within two years.

1. Introduction

Virtual Production (VP) has transformed traditional filmmaking [LWR25]. Instead of shooting actors on greenscreen and adding the environment months later in post, productions increasingly place actors inside LED volumes that display that environment in real time, a technique known as In-Camera Visual Effects (ICVFX). ICVFX is the most visible part of VP, but not all of it. VP can be understood as an umbrella term for techniques that let filmmakers interact with digital content as they would on a traditional shoot, including world capture (location and set scanning) and visualization (previs, virtual camera) [Netnd].

Nearly all of these techniques involve a digital 3D environment, for instance as a scene to plan shots in or as the background shown on the LED wall, and the quality demands on these environments rise the closer they get to the final image. For example, a previs scene may be barebones with a rough blocking using simple shapes such as boxes, but content that ends up in the final pixel, whether on an LED wall which must run in real time or composited behind actors, must be photoreal to be combined with shot footage. Building such environments with traditional modeling and photogrammetry pipelines is slow and expensive, making environment creation a central bottleneck of VP. 3D Gaussian Splatting (3DGS) [KKLD23] appears to address this bottleneck. It can reconstruct a photorealistic 3D scene that renders in real time from ordinary camera or drone footage, or even from rendered CG images. When compared to mesh-based photogrammetry, 3DGS is more forgiving: it handles reflective, semi-transparent and detailed surfaces better [HPRS25], and playback is fast enough for real-time rendering [KKLD23].

Since its introduction the industry has taken notice. Rodeo FX, for example, converted its Imperial Palace set from "Dune: Prophecy" into splats for real-time virtual scouting [Her25], a first study shows it outperforming traditional capture methods for virtual backdrops in realism [HPRS25], and interviews with filmmaking professionals confirm strong interest in Gaussian-based environments for previsualization [LWR25]. While the use of splats spreads quickly through scouting, previs, and asset creation, open problems in relighting [AAPB26, GGL*24, BZZ*24], dynamic range [ZLR*25], and moving content [BMG25] still stand between them and final-pixel use. What is known is scattered across technical papers, industry talks, and case studies.

No published work has yet mapped where splats stand in each phase of the VP pipeline (pre-production, production and post-production) or checked this against what practitioners actually do on set. This paper addresses that gap.

The research question is: At which phases of the VP pipeline is 3DGS currently used, which benefits and limitations do practitioners report, and why does adoption drop off toward final-pixel work? The scope is capture-based 3DGS (splats from real camera or drone footage). The approach combines a systematic mapping of academic and industry publications, sorted into a pipeline framework of pre-production, production, and post-production, with a small survey with expert VP/VFX practitioners that tests how far the literature matches current on-set practice. The contributions are:

- a phase-by-phase map of where 3DGS is used along the VP pipeline
- a overview of the benefits and limitations per phase
- a comparison of this literature view with current on-set practice, based on an survey with expert VP/VFX practitioners

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2. Related Work

3DGS is part of a group of methods called radiance fields. The best-known earlier one is NeRF, which stores a scene in a neural network, but renders new views too slowly for real-time use [MST*20,LWR25]. 3DGS instead stores the scene as millions of small 3D Gaussians that can be drawn in real time [KKLD23]. Photogrammetry creates textured meshes, which are still the standard in CGI pipelines, but it loses fine details and struggles with some reflective and transparent surfaces where splats do better [HPRS25].

Existing work on 3DGS in VP looks at one stage, such as backdrops [HPRS25] or previs [LWR25], describes one production [Her25], or tries to solve one problem, such as relighting [AAPB26]. None of them looks at the whole pipeline or compares the papers with practice, which is what this paper does.

3. Methodology

This study has two parts: a systematic mapping of publications on 3DGS in VP, sorted by pipeline phase, and a small practitioner survey that checks this literature picture against current practice in the industry.

3.1. Systematic mapping

The mapping follows the usual procedure for such studies [PFMM08]. As framework it uses the three classic production phases with finer stages inside them: pre-production (previs, techvis, digital scouting, environment/backdrop creation), production (final-pixel on the LED volume, greenscreen/virtual studio, engine integration), and post-production (compositing, relighting, cleanup, asset reuse).

Sources were collected through paper recommendations from fellow researchers and through web and database searches using "Gaussian Splatting" and "virtual production" as search terms with arXiv as the main source, complemented by publication venues such as ACM and IEEE. Collection ended in July 2026. A source is included if it (1) uses, evaluates, or directly extends 3DGS and (2) addresses a VP pipeline stage, either by explicitly placing itself in VP or film production, or by targeting a named VP concern such as scanning for capture, engine integration, relighting, or dynamic content. General 3DGS research without this framing, such as relighting methods [GGL*24, BZZ*24], serves as background only. The paper holds 13 unique sources from 2025 to 2026: 4 peer-reviewed papers, 3 preprints, and 6 talks, posters, demos, and case studies. Each source was sorted into every stage it deals with, so one source can show up in several stages. A stage is a main focus if the source's own work is about it: its method, its experiment, or its production case. A stage is only mentioned if the source names it as a motivation, a possible use, a limitation, or future work, or if it was just a tool to reach another goal. Relighting research counts as post-production, where it is applied, even though the same limitation also blocks final-pixel use.

3.2. Practitioner survey

To capture today's reality of 3DGS in virtual production, VP and VFX practitioners were invited to take part in a 5-minute online survey between July 14 and August 14, 2026. Participants were purposely sampled based on their professional background, requiring active roles in Virtual Production (VP) or VFX. Recruitment was conducted by posting an open call in Starting Pixel Community and by directly contacting verified industry professionals via LinkedIn. The survey collected responses from 13 professionals with experience levels ranging from under two years to over a decade. Since the sample is small, it is a convenience sample (n=13). For that reason, the results should not really be read as representative, but rather show a tendency. The survey asked for a mix of quick choices and general insights. Multiple-choice questions covered roles, experience, and their specific 3DGS workflows. Respondents also rated the tech's readiness across six production stages on a 1–5 scale, selected its biggest benefits and limitations, and guessed when it might become a standardized tool for final-pixel work. To go beyond the numbers, open questions invited them to share their go to capture tools, concrete success or failure cases and explain what it would take to fully trust 3DGS on set. While the sample size is small, the participants are highly qualified. Six of the 13 are VP Supervisors or Stage Leads (Figure 1), and five have already used 3DGS on a live project. Respondents are numbered R1 to R13 in order of submission. The complete questionnaire and its composition are provided in Appendix A.

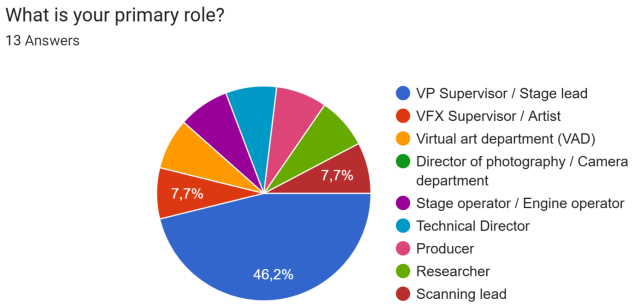


Figure 1: *Roles of the survey respondents ($n = 13$). Source: Google Forms.*

4. Results

4.1. Where splats are used

Table 1 shows where the 13 sources and the survey respondents use splats. In the literature, pre-production clearly dominates: 10 of the 13 sources focus on it, 8 of them on creating environments and backdrops, for example smartphone backdrops [HPRS25], scans of a historic building [Mir25], student captures of artworks [Ols25], assets made from archive material [ICM*25], scenes from a single image [SO26], scanning with several cameras at once [CSZ*26] and HDR captures [ZLR*25]. Previs and scouting are covered by one study with filmmakers [LWR25] and one production report [Her25].

Production comes up in 9 sources, but mostly on the side. The most common topic is getting splats into game engines (7 of these 9 sources), for example lighting CG objects inside a splat scene [TKSB25]. Only one source shows splats as final pixel on an LED wall during a real shoot [DB26]. Post-production shows up in 6 sources, mainly reusing captured assets and relighting and

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Stage	Main focus of	Also mentioned in	Survey
Pre-production			9
Previs	[LWR25]	[Her25]	5
Tech-vis	none	[LWR25]	1
Digital scouting	[Her25]	[LWR25]	6
Environment/backdrop creation	[CSZ*26, DB26, HPRS25, ICM*25, Mir25, Ols25, SO26, ZLR*25]	none	7
Production			6
Final pixel on the LED volume	[DB26]	[AAPB26, LWR25, Mir25, SO26]	6
Greenscreen/virtual studio	none	[HPRS25, ICM*25]	2
Engine integration	[Mir25, SO26, TKSB25]	[DB26, Her25, ICM*25, LWR25]	n/a
Post-production			3
Compositing	[AAPB26]	[Her25]	n/a
Relighting	[AAPB26]	[TKSB25]	n/a
Cleanup	none	[Mir25]	n/a
Asset reuse	[ICM*25]	[Her25, Ols25]	n/a

Table 1: *Where 3DGS is used along the VP pipeline. "Main focus of" lists the sources that are mainly about this stage, "Also mentioned in" the ones that only touch it. Survey lists how many of the 13 respondents used 3DGS at this stage (n/a = the survey had no answer option for it; post-production was asked as one option, "Post-production / compositing").*

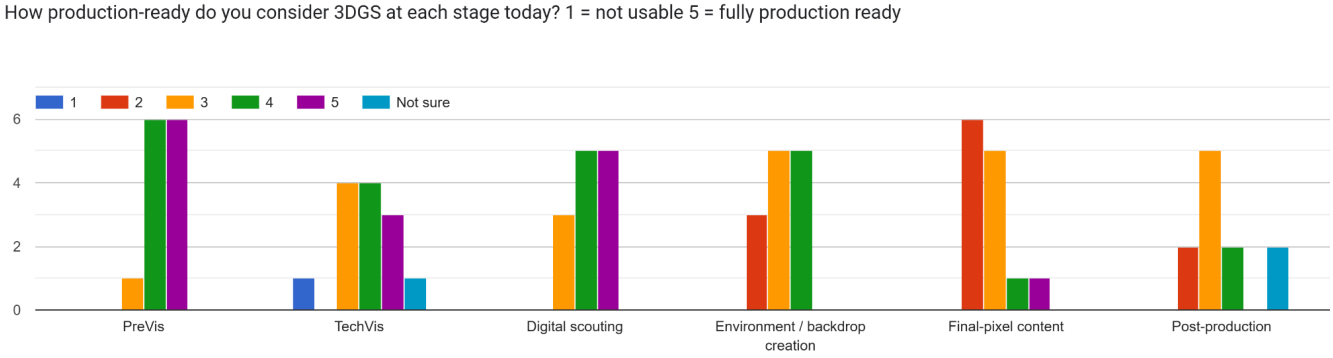


Figure 2: *How production-ready the respondents think 3DGS is at each stage, from 1 (not usable) to 5 (fully production ready). Chart from Google Forms, n = 13; "not sure" answers excluded from averages.*

compositing LED-volume footage [AAPB26]. Moving content is treated as a problem for all phases [BMG25].

In practice, the use is spread more evenly: respondents used splats most for environment creation (7 of 13), scouting (6) and on the LED volume (6), then for previs (5), post-production (3) and greenscreen (2). Twelve work mainly with real-world captures and four with AI-generated splats (more than one answer was possible). Postshot (9), Volinga (8) and Luma AI (7) are the most used tools.

4.2. How ready splats are

The readiness ratings go down along the pipeline (Figure 2). Previs scores highest (average 4.38 out of 5, 12 of 13 rate it 4 or 5), followed by scouting (4.15), tech-vis (3.67), environment creation (3.15), post-production (3.00) and final-pixel content (2.77), where 6 of 13 gave a 2. Asked directly if splats are ready today as final-pixel background on an LED wall, respondents are a bit more positive (average 3.46, 7 of 13 agree), and 9 of 13 rated this higher than the final-pixel stage. All 13 agree that 3DGS will be a standard part of VP within two years (average 4.46). The five respondents who used splats on a real production rate final pixel higher than those who only tested them (3.40 vs. 2.38).

4.3. Benefits and limitations

On the benefits, papers and practitioners agree (Table 2): fast and cheap capture (11 of 13 respondents, 9 of 13 sources) and realism (10 of 13 and 12 of 13) come first. However, on the limitations they do not agree. Respondents name relighting, editability, artifacts and moving content equally often (8 each), then dynamic range and color (5 each). The papers name file size and hardware most often (6), then moving content (5). Editability shows up in only 2 sources [LWR25, Mir25], and so does color management [Her25, LWR25].

The open answers make this clearer. Of 11 usable examples, 10 describe a success, 6 of them in pre-production. Both failures

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are about real-time performance in Unreal, once on a multi-node nDisplay setup (R1, R12). The two final-pixel successes came with conditions: splats worked for "mid and far distance environments" (R3) and for a warehouse with "easy shapes to relight" (R6). To trust splats for final pixel, respondents want better visual quality (8), editing tools (6), relighting (4), pipeline integration and standards (4) and more accurate color (3). One supervisor calls splats "sticky": quick to create, but slow to change afterwards (R5).

	Survey (13)	Papers (13)
Benefits		
Fast and cheap capture	11	9
Realism	10	12
Real-time playback	6	8
Easy to learn and use	4	4
Faster iteration with creatives	3	4
Works with simple devices	2	6
Limitations		
Relighting	8	4
Editability / art direction	8	2
Artifacts / visual quality	8	4
Moving content	8	5
Dynamic range (HDR)	5	2
Color accuracy	5	2
File size / hardware	4	6
Engine integration	4	4
Rights / legal	1	1
Capture effort	not asked	4

Table 2: *Benefits and limitations in the survey and in the papers. Survey: how many respondents picked the option (the limit of three picks was not enforced by the form). Papers: how many of the 13 sources mention it.*

5. Discussion

5.1. Why readiness drops toward final pixel

Papers and the survey participants tell the same story, splats work where a static, roughly correct image is enough. Respondents rate Splats most ready for previs (4.38) and digital scouting (4.15), and most of the successes they describe come from pre-production. At these stages baked-in light, a frozen scene and a few artifacts are fine, because nobody films the result. For final pixel they are not, the background has to match the light on set, change when the director wants something else, sometimes move, look clean through a cinema lens and work with the color pipeline [Her25, LWR25]. For each of these points there are only first solutions so far [AAPB26, BMG25, ZLR*25], and running splats smoothly in the engine is a practical problem on top [Mir25, SO26].

Still, final pixel is not a yes or no question. Six of 13 survey respondents already use splats on the LED volume. The one real case from the sources worked with a static practical set whose look should stay the same, and a camera that was locked off or only moved a little by hand [DB26]. Two survey participants describe the same pattern: splats worked for mid- and far distance environments (R3) and for a warehouse with simple shapes that were easy to relight (R6). This fits what respondents report for distant or simple scenes, and it explains why the LED background statement got better ratings than the final-pixel stage (3.46 vs. 2.77). So adoption does not stop at final pixel, it narrows down to the shots where these conditions are met.

5.2. What practice adds

The biggest difference is editability: 8 of 13 respondents count it among the biggest limitations, but only 2 of 13 sources mention it. The sources treat a splat as a reconstruction that should be rendered as exactly as possible, practitioners treat it like a set that someone will want to change. Because the look is baked in during capture, a fast start turns into a slow change later (R4, R5). Color management shows the same gap on a smaller scale (5 of 13 respondents vs. 2 of 13 sources). All 13 respondents nevertheless expect 3DGS to become a standard part of VP within two years, while rating final pixel lowest of all stages. Two explanations are supported by the study itself. First, respondents were recruited through a VP community and chose to answer a survey on 3DGS, so they are likely more optimistic than the field as a whole. Second, experience seems to raise confidence: the five respondents who used splats on a real production rate final pixel higher than those who only tested them (3.40 vs. 2.38). The expectation may therefore reflect what early adopters already see in their own work rather than a general industry view.

5.3. Limitations of this study

The biggest limitation of this study is the small number of sources. Because 3DGS is so new, there are only a few sources that included virtual production and were suitable for this research. The practitioner survey is small as well, it is a convenience sample of 13 people, about half of them are VP supervisors, and only 5 of them used splats on a real production. This is why the results show a tendency and not really a representative picture, and single answers matter a lot. For example, the only 5 for final pixel comes from a respondent who also reports a failed multi-node setup (R12). All in all the survey picks and mentions in papers are also not the same thing, so the Table 2 highlights which topics dominate, not how often they occur.

6. Future Work

A bigger survey with a more balanced distribution of roles and some one-on-one person interviews could lead to a better result. Two things found in this study also deserve closer attention: first the editability, which practitioners count as one of the biggest limitations but the literature barely addresses, and second the conditions under which splats can hold up as final pixel. The field is developing at a rapid rate, so repeating this study in one or two years could reveal a completely different picture of how Gaussian splats are used in virtual production.

7. Conclusion

This paper asked where along the VP pipeline 3DGS is used, which benefits and limitations practitioners report, and why adoption drops

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off towards final pixel. In pre-production, splats are already established, most sources focus on it, and practitioners rate them ready for previs and scouting. In production, they reach the LED wall, but final pixel only works under certain conditions, and getting splats into the engine is a recurring problem. In post-production, relighting, compositing and asset reuse are just starting. Speed and realism drive adoption, while relighting, editability, artifacts and moving content hold it back.

Acknowledgements

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AI Contribution

I also used AI from Claude (Anthropic; claude-opus-5-5, in Cowork) as a help. It was used to assist with the initial literature mapping. Beyond that, the tool was used as a help to refine specific passages, resolve phrasing uncertainties, condense sentences, and ensure the consistent and precise use of technical terminology. It is important to note, that all numbers, and references presented are fully verified by me.

References

[AAPB26] AZZARELLI A., ANANTRASIRICHAI N., POLLOCK J., BULL D. R.: Relightable Gaussian splatting for virtual production using image-based illumination. arXiv preprint arXiv:2605.09024, 2026. doi:10.48550/arXiv.2605.09024. 1, 2, 3, 4

[BMG25] BELEMKOABGA D. S., MAUGEY T., GUILLEMOT C.: GS-Morph: Dynamic novel view synthesis via UDF-ARAP Gaussian splat morphing. In *European Conference on Visual Media Production (CVMP '25)* (2025), ACM. doi:10.1145/3756863.3769701. 1, 3, 4

[BZZ*24] BI Z., ZENG Y., ZENG C., PEI F., FENG X., ZHOU K., WU H.: GS^3: Efficient relighting with triple Gaussian splatting. In *SIGGRAPH Asia 2024 Conference Papers* (2024), ACM, pp. 1–12. doi:10.1145/3680528.3687576. 1, 2

[CSZ*26] CAO Z., SHAO Q., ZHAI S., ZHANG J., NGUYEN A., HUANG B.: MAGS-SLAM: Monocular multi-agent Gaussian splatting SLAM for geometrically and photometrically consistent reconstruction. arXiv preprint arXiv:2605.10760, 2026. doi:10.48550/arXiv.2605.10760. 2, 3

[DB26] DÜNSER J., BLÖNNIGEN J.: Gaussian splatting in virtual production: How we put a human-sized can into a childhood bedroom. ACM SIGGRAPH Blog, https://blog.siggraph.org/2026/07/gaussian-splatting-in-virtual-production-how-we-put-a-human-sized-can-into-a-childhood-bedroom.html/, 2026. Posted July 8, 2026. 2, 3, 4

[GGL*24] GAO J., GU C., LIN Y., LI Z., ZHU H., CAO X., ZHANG L., YAO Y.: Relightable 3D Gaussians: Realistic point cloud relighting with BRDF decomposition and ray tracing. In *Computer Vision (ECCV 2024)* (2024), Springer, pp. 73–89. doi:10.1007/978-3-031-72995-9_5. 1, 2

[Her25] HERY J.: The environments of Dune: Prophecy through the Gaussian splat. In *SIGGRAPH Talks '25: Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks* (2025), ACM. doi:10.1145/3721239.3734124. 1, 2, 3, 4

[HPRS25] HASLBAUER P., PULLEN M., REICHHERZER C., SMOLIC A.: Gaussian splatting vs. classical photogrammetry: A comparison for virtual backdrops. In *2025 17th International Conference on Quality of Multimedia Experience (QoMEX)* (2025), IEEE. https://ieeexplore.ieee.org/document/11219987. 1, 2, 3

[ICM*25] IACOVIELLO R., CIPRIAN A., MESSINA A., MONTAGNUOLO M., ZAPPIA D.: XReco platform and RAI news media demonstrator. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)* (2025), ACM, p. 14361. doi:10.1145/3746027.3761840. 2, 3

[KKLD23] KERBL B., KOPANAS G., LEIMKÜHLER T., DRETTAKIS G.: 3D Gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics 42*, 4 (2023), 139:1–139:14. doi:10.1145/3592433. 1, 2

[LWR25] LEININGER P., WEBER C. J., ROTHE S.: Understanding creative potential and use cases of AI-generated environments for virtual film productions: Insights from industry professionals. In *Proceedings of the 2025 ACM International Conference on Interactive Media Experiences (IMX '25)* (2025), ACM, pp. 60–78. doi:10.1145/3706370.3727853. 1, 2, 3, 4

[Mir25] MIROCHA L.: Beyond digital twins: 3D Gaussian splatting, game engines and crossmedia cultural heritage representations. In *SIGGRAPH Talks '25: Special Interest Group on Computer Graphics and Interactive Techniques Conference Talks* (2025), ACM. doi:10.1145/3721239.3734094. 2, 3, 4

[MST*20] MILDENHALL B., SRINIVASAN P. P., TANCIK M., BARRON J. T., RAMAMOORTHI R., NG R.: NeRF: Representing scenes as neural radiance fields for view synthesis. In *Computer Vision (ECCV 2020)* (2020), Springer, pp. 405–421. doi:10.1007/978-3-030-58452-8_24. 2

[Netnd] NETFLIX, INC.: What is virtual production? Netflix Partner Help Center, https://partnerhelp.netflixstudios.com/hc/en-us/articles/1500002552642-What-is-Virtual-Production, n.d. 1

[Ols25] OLSEN C. K.: Virtualizing the Stanley: Gaussian splatting as educational gateway to advanced CGI techniques. In *SIGGRAPH Educators Forum '25* (2025), ACM. doi:10.1145/3721242.3734015. 2, 3

[PFMM08] PETERSEN K., FELDT R., MUJTABA S., MATTSSON M.: Systematic mapping studies in software engineering. In *Proceedings of the 12th International Conference on Evaluation and Assessment in Software Engineering (EASE '08)* (2008), BCS Learning & Development, pp. 68–77. doi:10.14236/ewic/EASE2008.8. 2

[SO26] SUN X., O'SULLIVAN C.: Still2Scene: Hybrid Gaussian environments for virtual production. In *Eurographics 2026 Posters* (2026), Gerrits T., Teschner M., (Eds.), The Eurographics Association. doi:10.2312/egp.20261012. 2, 3, 4

[TKSB25] TEUBER M., KUNERT C., SCHWANDT T., BROLL W.: Geometry-based light probe placement for realtime lighting in Gaussian splatting environments. *The Visual Computer 41* (2025), 11783–11795. doi:10.1007/s00371-025-04124-z. 2, 3

[ZLR*25] ZHANG K., LI M., REN M., DENG J., ZHU X.: High dynamic range 3D Gaussian splatting via luminance-chromaticity decomposition. arXiv preprint arXiv:2511.12895, 2025. doi:10.48550/arXiv.2511.12895. 1, 2, 3, 4

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Appendix A: Survey Questionnaire

The following questions were presented to practitioners in the online survey.

Experience

- What is your primary role?
- How many years have you worked in Virtual production or VFX?
- What is your experience with 3D Gaussian Splatting?

Pipeline & Workflows

- At which stages of your pipeline have you or your team used 3DGS?
- Are the 3D Gaussian Splats you work with primarily...? (AI-generated vs. capture)
- Which Capture methods do you use to create your 3DGS assets?
- Which, if any, specific plugins or software suites have you utilized for 3DGS?

Ratings

- How production-ready do you consider 3DGS at each stage today? (Rated 1 = not usable, 5 = fully production ready for: Pre-Vis, TechVis, Digital scouting, Environment / backdrop creation, Final-pixel content, Post-production)
- What are the biggest benefits of 3DGS in your experience or expectation? (pick up to 3)
- What are the biggest limitations holding 3DGS back? (pick up to 3)
- Rate your agreement: "3DGS is ready today to be used as final-pixel background content on an LED wall."
- Rate your agreement: "Within two years, 3DGS will be a standard part of virtual production workflows."

Open-Ended Qualitative Feedback

- Briefly describe one concrete case where 3DGS worked well, or failed, on a project.
- What would need to change for you to trust 3DGS for final-pixel work?
- What do you like or dislike about working with Splats? (optional)

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